

```
!pip install spacy transformers torch pandas datasets sequeval -q
!python -m spacy download en_core_web_sm
```

```

43.6/43.6 kB 3.0 MB/s eta 0:00:00
Preparing metadata (setup.py) ... done
Building wheel for sequeval (setup.py) ... done
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en\_core\_web\_sm-3.8.0/en\_core\_web\_sm-3.8.0.whl
12.8/12.8 MB 64.5 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
# spaCy for NLP processing and pretrained NER
import spacy
from spacy import displacy

# Hugging Face transformers for pretrained transformer models
from transformers import pipeline, AutoTokenizer, AutoModelForTokenClassification

# PyTorch backend (required by transformers)
import torch

# Pandas for structured table output
import pandas as pd

# For evaluation metrics
from datasets import load_dataset
from sequeval.metrics import classification_report, f1_score
```

```
# Load English pretrained model
nlp = spacy.load("en_core_web_sm")
```

```
sentences = [
    "Apple CEO Tim Cook announced the new iPhone 15 in California.",
    "Lionel Messi scored a goal for Inter Miami in the MLS final.",
    "The United Nations held a climate summit in New York.",
    "Elon Musk's SpaceX launched a rocket from Texas.",
    "Microsoft invested $10 billion in OpenAI."
]
```

```
print("----- spaCy NER Results -----\\n")

spacy_results = []

for sentence in sentences:
    doc = nlp(sentence)
    print(f"Sentence: {sentence}")

    for ent in doc.ents:
        print(f"Entity: {ent.text} | Label: {ent.label_}")
        spacy_results.append([sentence, ent.text, ent.label_])

    print("-" * 60)
```

```
----- spaCy NER Results -----
```

```
Sentence: Apple CEO Tim Cook announced the new iPhone 15 in California.
Entity: Apple | Label: ORG
Entity: Tim Cook | Label: PERSON
```

```
Entity: 15 | Label: CARDINAL
Entity: California | Label: GPE
-----
Sentence: Lionel Messi scored a goal for Inter Miami in the MLS final.
Entity: Lionel Messi | Label: PERSON
Entity: MLS | Label: ORG
-----
Sentence: The United Nations held a climate summit in New York.
Entity: The United Nations | Label: ORG
Entity: New York | Label: GPE
-----
Sentence: Elon Musk's SpaceX launched a rocket from Texas.
Entity: Elon Musk's | Label: PERSON
Entity: SpaceX | Label: PERSON
Entity: Texas | Label: GPE
-----
Sentence: Microsoft invested $10 billion in OpenAI.
Entity: Microsoft | Label: ORG
Entity: $10 billion | Label: MONEY
Entity: OpenAI | Label: GPE
-----
```

```
df_spacy = pd.DataFrame(spacy_results, columns=["Sentence", "Entity", "Label"])
df_spacy
```

	Sentence	Entity	Label	
0	Apple CEO Tim Cook announced the new iPhone 15...	Apple	ORG	
1	Apple CEO Tim Cook announced the new iPhone 15...	Tim Cook	PERSON	
2	Apple CEO Tim Cook announced the new iPhone 15...	15	CARDINAL	
3	Apple CEO Tim Cook announced the new iPhone 15...	California	GPE	
4	Lionel Messi scored a goal for Inter Miami in ...	Lionel Messi	PERSON	
5	Lionel Messi scored a goal for Inter Miami in ...	MLS	ORG	
6	The United Nations held a climate summit in Ne...	The United Nations	ORG	
7	The United Nations held a climate summit in Ne...	New York	GPE	
8	Elon Musk's SpaceX launched a rocket from Texas.	Elon Musk's	PERSON	
9	Elon Musk's SpaceX launched a rocket from Texas.	SpaceX	PERSON	
10	Elon Musk's SpaceX launched a rocket from Texas.	Texas	GPE	
11	Microsoft invested \$10 billion in OpenAI.	Microsoft	ORG	
12	Microsoft invested \$10 billion in OpenAI.	\$10 billion	MONEY	
13	Microsoft invested \$10 billion in OpenAI.	OpenAI	GPE	

Next steps:

[Generate code with df_spacy](#)

[New interactive sheet](#)

```
doc = nlp(sentences[0])
displacy.render(doc, style="ent", jupyter=True)
```

Apple **ORG** CEO Tim Cook **PERSON** announced the new iPhone 15 **CARDINAL** in California **GPE** .

```
model_name = "dbmdz/bert-large-cased-finetuned-conll03-english"

tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForTokenClassification.from_pretrained(model_name)

ner_pipeline = pipeline("ner", model=model, tokenizer=tokenizer)
```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens)
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
config.json: 100% 998/998 [00:00<00:00, 50.5kB/s]

tokenizer_config.json: 100% 60.0/60.0 [00:00<00:00, 4.33kB/s]

vocab.txt: 213k/? [00:00<00:00, 11.0MB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable faster authentication.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable faster authentication.
model.safetensors: 100% 1.33G/1.33G [00:14<00:00, 128MB/s]

Loading weights: 100% 391/391 [00:00<00:00, 573.52it/s, Materializing param=classifier.weight]
BertForTokenClassification LOAD REPORT from: dbmdz/bert-large-cased-finetuned-conll103-english
Key | Status | 
-----+-----+---
bert.pooler.dense.weight | UNEXPECTED | 
bert.pooler.dense.bias | UNEXPECTED | 

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expected

```

```

print("----- Hugging Face NER Results -----\\n")

hf_results = []

for sentence in sentences:
    outputs = ner_pipeline(sentence)

    print(f"Sentence: {sentence}")
    for ent in outputs:
        print(f"Word: {ent['word']} | Label: {ent['entity']} | Score: {ent['score']:.4f}")

    print("-" * 60)

```

```

----- Hugging Face NER Results -----

Sentence: Apple CEO Tim Cook announced the new iPhone 15 in California.
Word: Apple | Label: I-ORG | Score: 0.9973
Word: Tim | Label: I-PER | Score: 0.9995
Word: Cook | Label: I-PER | Score: 0.9997
Word: iPhone | Label: I-MISC | Score: 0.9949
Word: 15 | Label: I-MISC | Score: 0.9947
Word: California | Label: I-LOC | Score: 0.9997
-----

Sentence: Lionel Messi scored a goal for Inter Miami in the MLS final.
Word: Lionel | Label: I-PER | Score: 0.9991
Word: Me | Label: I-PER | Score: 0.9992
Word: ##ssi | Label: I-PER | Score: 0.9982
Word: Inter | Label: I-ORG | Score: 0.9989
Word: Miami | Label: I-ORG | Score: 0.9645
Word: MLS | Label: I-ORG | Score: 0.9215
-----

Sentence: The United Nations held a climate summit in New York.
Word: United | Label: I-ORG | Score: 0.9994
Word: Nations | Label: I-ORG | Score: 0.9994
Word: New | Label: I-LOC | Score: 0.9994
Word: York | Label: I-LOC | Score: 0.9994
-----

Sentence: Elon Musk's SpaceX launched a rocket from Texas.
Word: El | Label: I-PER | Score: 0.9996
Word: ##on | Label: I-PER | Score: 0.9991
Word: Mu | Label: I-PER | Score: 0.9993
Word: ##sk | Label: I-PER | Score: 0.9979
Word: Space | Label: I-ORG | Score: 0.9995

```

Word: ##X | Label: I-ORG | Score: 0.9989
 Word: Texas | Label: I-LOC | Score: 0.9996

 Sentence: Microsoft invested \$10 billion in OpenAI.

Word: Microsoft | Label: I-ORG | Score: 0.9994
 Word: Open | Label: I-ORG | Score: 0.4667
 Word: ##A | Label: I-ORG | Score: 0.8100
 Word: ##I | Label: I-ORG | Score: 0.8666

```
clean_results = []

for sentence in sentences:
    outputs = ner_pipeline(sentence)

    current_entity = ""
    current_label = ""
    current_score = 0

    for ent in outputs:
        word = ent['word'].replace("##", "")
        label = ent['entity'].split("-")[-1]

        if "B-" in ent['entity']:
            if current_entity:
                clean_results.append([sentence, current_entity, current_label, current_score])
                current_entity = word
                current_label = label
                current_score = ent['score']
            else:
                current_entity += " " + word

        if current_entity:
            clean_results.append([sentence, current_entity, current_label, current_score])

df_hf = pd.DataFrame(clean_results,
                     columns=["Sentence", "Entity", "Label", "Confidence Score"])

df_hf
```

	Sentence	Entity	Label	Confidence Score
0	Apple CEO Tim Cook announced the new iPhone 15...	Apple Tim Cook iPhone 15	California	0
1	Lionel Messi scored a goal for Inter Miami in ...	Lionel Me ssi Inter Miami MLS		0
2	The United Nations held a climate summit in Ne...	United Nations New York		0
3	Elon Musk's SpaceX launched a rocket from Texas.	El on Mu sk Space X Texas		0

Next steps: [Generate code with df_hf](#) [New interactive sheet](#)

```
comparison = pd.DataFrame({
    "Feature": ["Model Type", "Speed", "Accuracy", "Context Handling", "Confidence Score"],
    "spaCy": ["Statistical model", "Very Fast", "Moderate", "Limited context", "No"],
    "Hugging Face": ["Transformer (BERT)", "Slower", "High", "Full sentence context", "Yes"]
})

comparison
```

	Feature	spaCy	Hugging Face	
0	Model Type	Statistical model	Transformer (BERT)	
1	Speed	Very Fast	Slower	
2	Accuracy	Moderate	High	
3	Context Handling	Limited context	Full sentence context	
4	Confidence Score	No	Yes	

Next steps:

[Generate code with comparison](#)[New interactive sheet](#)